AI-Powered Lung Disease Detection: An Explainable Dual-Modality System for Chest X-Ray and CT Screening

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***Abstract*—Most deep-learning tools for chest imaging handle one modality, return a label without an explanation, and confidently label any image they are given. We present a web-based system in which transfer-learned DenseNet-121 networks classify chest radiographs as COVID-19, Normal, Pneumonia or Tuberculosis and computed tomography (CT) slices as Benign, Malignant or Normal. Each upload first passes image validation, a sanity gate that rejects non-scans, a modality gate and a feature-space out-of-distribution detector; X-rays are also cropped to the lung fields by a U-Net. Each result carries a Grad-CAM heatmap, a non-diagnostic report from a large language model and a scoped chat assistant, in an application with verified accounts, scan history and rate limiting, deployed over HTTPS on a cloud server. Data audits removed near-duplicate CT slices that had inflated test accuracy to 100% and burned-in artifacts from tuberculosis images. Internal validation accuracy is 96.5% for chest X-ray and 99.0% for lung CT, with held-out test accuracy of 96.57% and 98.06%, and after a train/serve preprocessing fix the deployed X-ray route delivers 91.05%. On new images from outside the training sources, sent through the deployed routes, accuracy is 93.9% for X-ray (31 of 33 images accepted by the guards) and 91.9% for CT.**

***Index Terms*—chest X-ray, computed tomography, explainable AI, external validation, Grad-CAM, out-of-distribution detection, transfer learning**

I. Introduction

Radiographs and computed tomography (CT) remain the main imaging routes by which clinicians identify tuberculosis, pneumonia, COVID-19 and lung cancer. Automated screening support matters especially in India, which carries about a quarter of the world's tuberculosis cases [1], where 80–85% of lung-cancer patients present with incurable disease [2], and where there is roughly one radiologist per 100,000 people [3]. Convolutional neural networks (CNNs) can match radiologists on narrowly defined chest-imaging tasks [4], [5], which has made automated triage an active area of applied research.

Three practical gaps limit how such models can be used. First, most systems handle one modality and give no explanation, so a clinician receives a label they cannot inspect. Second, a discriminative classifier assigns one of its known labels to *any* input, including photographs, screenshots and scans of the wrong type; in our testing, a CT model labeled a screenshot "Malignant" with 93% confidence. Third, accuracy is usually reported offline on test images from the training sources, which says little about what a deployed system delivers.

Problem statement. The goal is a single system that assigns a radiograph to one of COVID-19, Normal, Pneumonia or Tuberculosis and a CT slice to one of Benign, Malignant or Normal, explains every prediction, refuses images it was not trained to handle, and is delivered as a secure multi-user web application whose accuracy is verified on the deployed code path and on external data.

This paper presents such a system. Its contributions are:

1) A dual-modality pipeline: DenseNet-121 [6] classifiers with chest-radiograph pretraining for X-ray [7] and ImageNet pretraining for CT, Grad-CAM heatmaps [8], and a safety-constrained large language model (LLM) report and chat assistant.
2) An inference-time guard layer (image checks, a per-modality sanity gate, a modality gate and a feature-space out-of-distribution (OOD) detector) and a start-up check that refuses to serve a checkpoint whose validation metric cannot be reproduced.
3) Data-quality audits (artifact cleanup, detection of near-duplicate leakage with cluster-aware resplitting, exclusion of a protocol mismatch, and a source-confound probe) that materially changed the reported results.
4) A live-serving audit that found and largely closed a 12–19-point train/serve accuracy gap, and an external evaluation on new images through the deployed routes (93.9% X-ray, 91.9% CT).
5) A complete web application (FastAPI, React, verified accounts, scan history, rate limiting) deployed in Docker behind an HTTPS reverse proxy on a cloud server.

II. Related Work

Chest X-ray classification. CheXNet [4] showed that a DenseNet-121 trained on ChestX-ray14 detects pneumonia at a level comparable to radiologists, CheXpert [5] provided a large multi-label benchmark, and COVID-Net [9] applied a similar approach to COVID-19. On the datasets used here, Chowdhury et al. [10] screened viral and COVID-19 pneumonia, and Rahman et al. [11] detected tuberculosis after lung segmentation, a design close to our U-Net crop. Mirugwe et al. [12] compared six ImageNet-pretrained CNNs on the same tuberculosis database and found that the simplest, VGG16, gave the best balance of accuracy and computational cost.

TABLE I
POSITIONING AGAINST RELATED WORK

System	Two modalities	Input guards	Leakage audit	External test
CheXNet [4]	No	No	–	No
COVID-Net [9]	No	No	–	No
Rahman et al. [11]	No	No	–	No
Benamara et al. [14]	No	No	No	No
This work	Yes	Yes	Yes	Yes

TABLE II
FINAL DATA SPLITS AFTER CLEANING

Modality	Class	Train	Val	Test	Total
X-ray	COVID-19	2,856	357	357	3,570
	Normal	10,949	1,369	1,368	13,686
	Pneumonia	1,070	134	134	1,338
	Tuberculosis	498	62	63	623
CT	Benign	96	12	12	120
	Malignant	400	50	50	500
	Normal	333	42	41	416

CT classification. CNNs are widely used to classify pulmonary nodules and lung cancer on CT [13]; on the CT collection used here, Benamara et al. [14] reported 99.54% accuracy with multi-level fine-tuning of a pretrained VGG16. Such studies typically use small single-institution datasets and rarely report a near-duplicate audit of their splits or an external test.

Unfamiliar inputs and reports. Hendrycks and Gimpel [15] showed that softmax confidence is an imperfect signal for detecting unfamiliar inputs, which our testing reproduced. Liu et al. [16] survey automatic report generation; our report layer instead uses a narrowly constrained prompt with a deterministic fallback.

As Table I shows, none of these systems combines two modalities with input guards, a leakage audit and an external test.

III. DATA AND DATA-QUALITY AUDITS

Datasets. Radiographs come from two public collections: the first [17] supplies the COVID-19, Normal and viral Pneumonia classes with a lung mask per image (its Lung Opacity class is not used), and the second [18] supplies Normal and Tuberculosis images. CT slices come from IQ-OTH/NCCD [19] (1,097 slices from 110 patients at one institution, exported as per-class JPEGs without patient IDs). The two X-ray sources were merged and 59 exact duplicates removed by SHA-256 hashing. Table II gives the final splits (seed 42); Normal outnumbers Tuberculosis about 22 times, and Malignant outnumbers Benign about 4.2 times. Auxiliary data comprise 21,165 lung image–mask pairs for the U-Net and curated photographs, documents and screenshots as negatives for the sanity gates.

Class sources. The viral Pneumonia images of [17] come from the pediatric collection of Kermany et al. [20] (patients aged one to five years at one hospital in Guangzhou), as do 1,341 of its Normal images, whereas the remaining Normal images and the Tuberculosis images are mostly adult radiographs. Every Pneumonia image in this work is therefore pediatric, and COVID-19 and Tuberculosis each come from a single database.

Burned-in artifacts. A scan of all 4,200 images of the TB source found a color tint and/or a white redaction box in 53.1% of Tuberculosis images but only 0.37% of Normal images, a class-correlated shortcut. Both classes were cleaned uniformly: tinted images were desaturated, small boxes inpainted and images with large boxes excluded (72 of 700 TB images).

CT leakage. The first CT model, trained on a random per-image split, scored 100% on test. Perceptual hashing with structural-similarity (SSIM) confirmation [21] found 111 clusters of near-duplicate slices (287 pairs); 47 clusters spanned splits and 35 of 109 test images had a near-duplicate in training. Treating clusters as patient proxies, the data were resplit so that each cluster lies in one split (zero cross-split leaks on re-audit). A further audit found 61 Malignant slices from a different acquisition protocol (no visible scanner table, non-standard resolution) that were separable on protocol alone; they were excluded, leaving 1,036 slices.

Source confound. In the X-ray data, COVID-19 comes only from one database and Tuberculosis only from the other. Within the Normal class, which both databases contribute to, a logistic-regression probe on 32×32 thumbnails recovered the source database at 97.1% accuracy (AUC 0.996) on 3,495 images per source.

IV. PROPOSED SYSTEM

A. Overview

The system is a three-tier web application: a React 19 single-page application; a FastAPI back end that loads six PyTorch models once at start-up (two sanity gates, a modality gate, two disease classifiers and a U-Net) and shares them across requests; and a SQLite database, with the Groq LLM API, Gmail SMTP and Google OAuth as external services. Fig. 1 shows the per-request prediction pipeline. Separate routes, `/predict/xray` and `/predict/ct`, make model selection a routing decision rather than a branch on a user-supplied field.

B. Disease Classifiers

Both classifiers use DenseNet-121 [6], whose feature reuse suits fine-tuning on datasets far smaller than ImageNet and whose roughly 8M parameters keep CPU inference practical. The X-ray model uses the TorchXRayVision (XRV) `densenet121-res224-all` weights [7], pretrained on a large aggregate of chest radiographs, with a new four-class head and the library's single-channel preprocessing; it replaced an earlier ImageNet-pretrained model. The CT model uses ImageNet weights [22] with a new three-class head. Both are trained for 15 epochs (batch 32, seed 42): two epochs with the backbone frozen using Adam [23] at 10^{-3}, then full fine-tuning at 10^{-4}, with inverse-frequency class weights $w_c = N/(Kn_c)$ in the cross-entropy loss and flip,

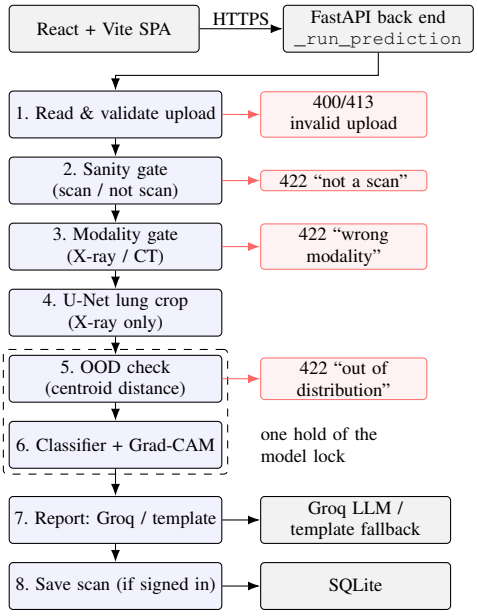


Fig. 1. Architecture and per-request prediction pipeline, top to bottom in execution order. Stages 1, 2, 3 and 5 can reject a request before the disease classifier runs. Both gates see the uncropped upload; only accepted X-rays are lung-cropped. Stages 5–6 run inside one hold of the classifier's lock.

$\pm 10°$ rotation and ± 0.1 brightness/contrast augmentation. The checkpoint with the best validation accuracy is kept (X-ray 96.46% at epoch 12; CT 99.04% at epoch 6).

C. Sanity Gate and Modality Gate

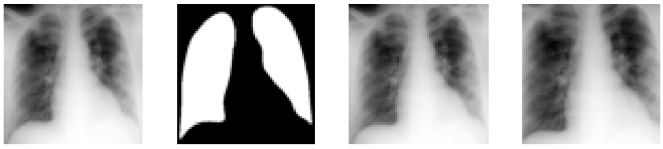
Both gates are ResNet-18 [24] binary classifiers. The two sanity gates (one per modality) separate real scans from photographs, documents and screenshots (X-ray: 15,373 positives vs. 7,460 negatives; CT: 829 vs. 3,000), reaching 99.96% and 100% validation accuracy. They run first because the modality gate is a closed-set classifier with no "neither" option: in testing, a random photograph was scored above 92% "X-ray" by the modality gate and also passed the OOD check. The modality gate, trained on the training splits relabeled by modality, reaches 100% validation accuracy; a request is rejected if it predicts the other modality or if its confidence is below 0.90. It was added because the OOD detector did not catch a real CT slice sent to the X-ray route.

D. Out-of-Distribution Detector

The detector works in the classifier's own penultimate (post-pooling) feature space $\phi(\cdot)$. For each modality, the training set is embedded offline and the centroid and threshold are stored:

$$\mu = \frac{1}{N}\sum_{i=1}^{N}\phi(x_i), \qquad \tau = Q_{0.99}\big(\{\|\phi(x_i)-\mu\|_2\}_{i=1}^{N}\big). \quad (1)$$

An input is rejected if $\|\phi(x)-\mu\|_2 > \tau$, i.e., if it is farther from the training data than 99% of training images. The thresholds are 8.72 for X-ray (recomputed after the backbone change) and 25.62 for CT.



(a) raw input (b) U-Net mask (c) seg. crop (d) border crop

Fig. 2. Lung-crop pipeline on a real test X-ray: (a) input; (b) U-Net mask; (c) segmentation crop; (d) border-crop fallback.

E. U-Net Lung Crop

The X-ray classifier was trained on lung-cropped images, and serving raw uploads changed its predictions (a genuine COVID-19 image was classified as Tuberculosis at 98.6% confidence). Accepted X-rays are therefore segmented by a 4-level U-Net [25] (32 base channels) at 256×256; the image is cropped to the mask's bounding box (4-pixel padding) and squash-resized to 224×224, as in training. If the mask is empty or its box covers less than 5% or more than 98% of the frame, a uniform 8% border crop is used instead (Fig. 2). The U-Net checkpoint was trained outside the project repository; its architecture was reconstructed from the checkpoint and confirmed by reproducing its validation Dice (Section V-B).

F. Explanation, Report and Chat

Grad-CAM [8] weights the last convolutional block's feature maps A^k by the spatially averaged gradients of the class score y^c, $\alpha_k^c = \frac{1}{Z}\sum_{i,j}\partial y^c/\partial A_{ij}^k$, and returns $\mathrm{ReLU}(\sum_k \alpha_k^c A^k)$, upsampled and alpha-blended (0.45) over the input; a new hook is registered per request. A low-confidence warning is attached when the top probability is below 0.50 (X-ray) or 0.55 (CT). The Groq-hosted `openai/gpt-oss-120b` model writes a two-to-three-sentence explanation from the class, confidence and probabilities under a prompt that forbids diagnosis, treatment advice and speculation; a fixed disclaimer is appended, and a template with the same disclaimer is used if the call fails. The chat assistant uses the same model with a system prompt restricted to the tool and the current result; it declines medical advice and off-topic requests, sees the last ten turns and accepts messages of up to 2,000 characters.

V. Implementation and Testing

A. Prediction Pipeline

Algorithm 1 gives `_run_prediction()`, shared by both prediction routes. The OOD check and the classification with Grad-CAM run inside one continuous hold of a per-model lock: FastAPI runs synchronous handlers in a thread pool, and Grad-CAM's forward hooks on a shared model let an overlapping request observe another request's hooks, a failure seen during development.

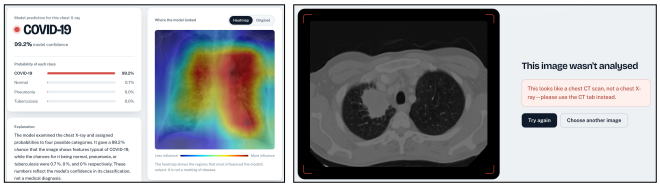
B. Model Loading and Reproduction Check

At start-up the server loads all six checkpoints, the two OOD statistics files and the U-Net once. Each checkpoint stores its claimed validation metric, which a reproduction

Algorithm 1 Prediction pipeline (`_run_prediction`)

Input: upload f, route modality m, optional user u
Output: class, probabilities, heatmap, report; or HTTP error

if u signed in **and** unverified **then return** 403
$I \leftarrow$ decode f ▷ 413 if >10 MB; 400 if invalid
if gray std < 10 **or** color spread > 20 **then return** 400
$x \leftarrow 224{\times}224$ ImageNet-normalized copy of I
if $\text{Sanity}_m(x) \neq$ scan **then return** 422
$(\hat{m}, p) \leftarrow \text{ModalityGate}(x)$
if $p < 0.90$ **or** $\hat{m} \neq m$ **then return** 422
if m = X-ray **then** $x \leftarrow \text{Prep}_{\text{XRV}}(\text{LungCrop}(I))$
acquire lock of Classifier_m
if $\|\phi_m(x) - \mu_m\|_2 > \tau_m$ **then** release; **return** 422
$(probs, heatmap) \leftarrow \text{Classifier}_m(x)$ with Grad-CAM
release lock; add warning if $\max(probs) < c_m$
$report \leftarrow$ LLM report, or template on failure
if u signed in **then** save scan for u
return result



(a) X-ray result (b) Guard rejection

Fig. 3. Screens of the deployed application: (a) COVID-19 prediction for a chest X-ray with class-probability bars, Grad-CAM heatmap and plain-language explanation; (b) a CT slice uploaded on the X-ray tab, refused by the modality gate with the reason.

check recomputes on the validation split with the serving preprocessing; in the default strict mode the server refuses to start if any metric differs by more than 0.1 percentage point (accuracy) or 1 point Dice / 2 points IoU (U-Net). In a verified start, all five classifiers and gates reproduced their validation accuracy exactly (X-ray 0.9646, CT 0.9904, modality gate 1.0000, sanity gates 0.9996 and 1.0000), and the U-Net reproduced Dice 0.983 and IoU 0.967 against 0.989 and 0.979 claimed. The check caught a real checkpoint/data mismatch during development, which is why it fails closed. It needs the validation data and takes two to three minutes per model on a CPU, so the hosted container, which does not ship the data, disables it with an environment switch, and the strict check is run on the development machine instead.

C. Application, Security and Deployment

The back end exposes fifteen routes (two prediction, nine authentication, chat, scan history and two status routes). Passwords are hashed with bcrypt, sessions are HS256 JSON Web Tokens valid for seven days, and email-verification (24 h) and password-reset (1 h) links are single-use tokens sent over Gmail SMTP. Google sign-in uses a signed `state` bound to a nonce cookie, email links and redirects follow the caller's origin only if it is allow-listed, and users with an unverified email receive HTTP 403 from prediction and chat. Per-client rate limits (20/min for prediction, chat and login; 30/h for sign-up; 5/h for password reset) use the real client address, trusting `X-Forwarded-For` only from a configured proxy. Each saved scan keeps its probabilities, heatmap and report, so the history page reproduces what the user saw. The React front end steps through a four-stage checklist while a request runs, then shows the class, confidence, per-class probabilities, the heatmap (with a toggle to the original image) and the explanation, labeled as written by a language model, beside a chat panel; a rejection is shown with its reason (Fig. 3).

The public deployment runs on one AWS EC2 instance (t3.small: 2 vCPUs, 2 GB RAM, 4 GB swap) under Docker Compose. A multi-stage Dockerfile builds the front end and installs CPU-only PyTorch; one process in the container serves the React build at `/` and the API under `/api`, so the site is a single origin. A Caddy container terminates HTTPS with an automatically renewed Let's Encrypt certificate and is the only proxy whose `X-Forwarded-For` header the back end trusts; the database lives on a named volume that survives redeployment. For local demonstrations, an ngrok HTTPS tunnel is used instead.

D. Testing

The `pytest` suite (81 tests in seven files) starts the real application with all six models, a temporary database and mocked SMTP and LLM clients, and covers predictions on real images, history, the 403 for unverified users, invalid and wrong-modality uploads, the uncertain-modality guard, authentication, origin handling, OAuth state tampering, client-IP extraction behind both proxies, single-origin serving and chat limits; all tests pass. Over live HTTP, wrong-modality scans, photographs, screenshots, documents and blank, noise and corrupt files were all rejected with the intended status codes, and twelve simultaneous predictions matched their sequential results exactly. Mean end-to-end latency was 1.26 s (X-ray) and 1.29 s (CT) over eight sequential requests each on the development laptop.

VI. Results and Discussion

A. Held-out Test Performance

Table III and Fig. 4 report the classifiers on their held-out test splits. Validation accuracy is 96.5% for X-ray and 99.0% for CT, and validation and test agree (X-ray 96.46% vs. 96.57%, 95% Wilson interval 95.7–97.3%; CT 99.04% vs. 98.06%, 93.2–99.5%). The CT figure was obtained after the leakage fix of Section III; its only errors are two Benign slices predicted Normal, and with 12 Benign test images each error moves Benign recall by 8.3 points. These are in-distribution results.

B. Live-Serving Accuracy and the Crop Fix

Sending test images through the deployed X-ray route revealed live accuracy 12–19 points below the offline figure, with predictions drifting to COVID-19 (350 of 436 served

TABLE III
HELD-OUT TEST PERFORMANCE (OFFLINE CLASSIFIER)

	Class	Prec.	Recall	F1	n
X-ray (acc. 96.57%)	COVID-19	0.926	0.944	0.935	357
	Normal	0.982	0.972	0.977	1,368
	Pneumonia	0.913	0.940	0.927	134
	Tuberculosis	0.955	1.000	0.977	63
	Macro avg.	0.944	0.964	0.954	1,922
CT (acc. 98.06%)	Benign	1.000	0.833	0.909	12
	Malignant	1.000	1.000	1.000	50
	Normal	0.953	1.000	0.976	41
	Macro avg.	0.984	0.944	0.962	103

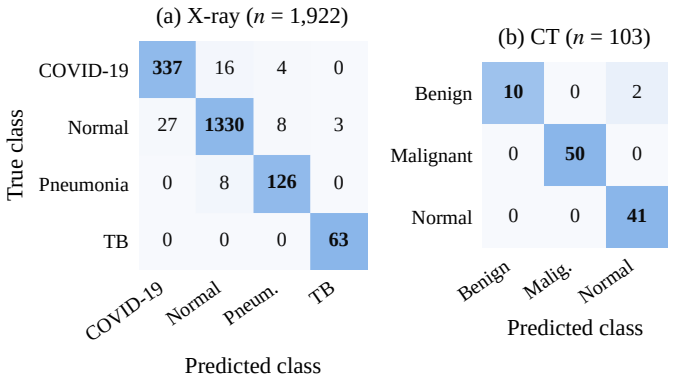


Fig. 4. Confusion matrices on the held-out test splits (counts; shading by row-normalized rate).

images of a class-balanced sample). The cause was the crop function, which could return a non-square box that the X-ray preprocessing then center-cropped, removing lung margins that every training image had kept. Ending every crop path in the training squash-resize closed most of the gap: on 240 class-balanced original uploads, live accuracy rose from 88.3% to 95.0% (offline 96.57%), and the share of served images predicted COVID-19 fell from 80% to 27%. The full 1,922-image test split through the live API with all guards active now gives 91.05% (95% CI 89.7–92.2%; rejections counted as errors; 91.67% over the 1,909 served images, 13 rejected by the OOD detector), and the CT route shows no loss (98.06% live). The remaining X-ray gap, mainly Normal predicted as COVID-19 (82/1,368), comes from a class-dependent crop difference in the training data (mask crops for COVID-source classes, border crops for TB-source classes). The reproduction check cannot detect such defects, because it replays the checkpoint on pre-processed validation data; only testing the real route did.

C. Internal vs. External Accuracy

To check the classifiers beyond their validation splits, new images from outside the training sources were sent through the deployed routes with every guard active. Table IV and Fig. 5 compare the internal (validation) accuracy with the accuracy on these external images.

TABLE IV
INTERNAL (VALIDATION) VS. EXTERNAL DETECTION ACCURACY

Modality	Internal (validation)	External (new images)
Chest X-ray	96.5%	93.9% (31 of 33)
Lung CT	99.0%	91.9%

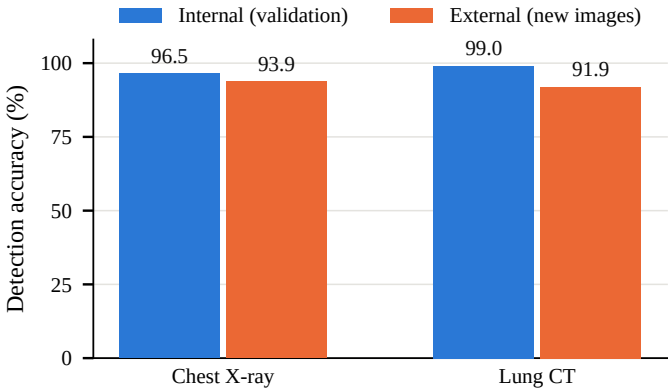


Fig. 5. Detection accuracy on the internal validation images and on new external images, for the chest X-ray and lung CT classifiers.

X-ray. Of the external radiographs, 33 were accepted by the guards and 31 were classified correctly, an accuracy of 93.9% (internal validation: 96.5%).

CT. On external CT slices the classifier reaches 91.9% (internal validation: 99.0%). Both classifiers therefore maintain high detection accuracy on images from outside their training data.

D. Limitations

The CT dataset is small and single-institution; the Pneumonia class is entirely pediatric, and COVID-19 and Tuberculosis each come from one database. The external sets are small and do not cover every class equally, so their accuracy figures have wide confidence intervals. The gates were trained on cropped images but receive uncropped uploads, and the OOD detector does not reject genuine chest images of untrained conditions (Lung Opacity X-rays were assigned to trained classes). The U-Net has no training script in the repository, and the hosted container runs without the reproduction check. Prediction and chat accept anonymous calls bounded only by rate limits, sessions cannot be revoked before expiry, rate limiting is per process and SQLite is single-writer, so a multi-server deployment would need changes; throughput under load was not measured.

VII. CONCLUSION AND FUTURE WORK

We presented a dual-modality system that classifies chest X-rays and CT slices with transfer-learned DenseNet-121 networks, explains every prediction with Grad-CAM and a non-diagnostic language-model report, refuses non-scans, wrong-modality and unfamiliar inputs, and can verify that every served checkpoint reproduces its validation metrics, inside a multi-user web application deployed over HTTPS. Internal

validation accuracy is 96.5% (X-ray) and 99.0% (CT), held-out test accuracy is 96.57% and 98.06%, and the deployed X-ray route delivers 91.05%. On new external images the system reaches 93.9% (X-ray) and 91.9% (CT). Auditing the data corrected an inflated CT accuracy, and testing the real route found and fixed a train/serve mismatch. Future work includes adult pneumonia images and several sources for every class, multi-institution CT data with patient identifiers, a larger external evaluation, one crop convention for all X-ray training data, gates trained on uncropped inputs, open-set rejection, confidence calibration, a clinical reader study, and production hardening (mandatory authentication, revocable sessions and continuous integration).

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